

A TWISTED EGUCHI–KAWAI SPECTRAL CURVE AT $SU(3)$ AND THE MODULAR FORM 24.2.a.a: AN EICHLER–SHIMURA IDENTIFICATION

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ABSTRACT. We observe that the one-loop propagator spectrum of the twisted Eguchi–Kawai (TEK) model for pure $SU(3)$ lattice gauge theory in $d = 4$ with the minimal symmetric twist, expanded around the clock-shift twist eaters, collapses to a four-point set $\{3, 6, 9, 12\}$. The associated hyperelliptic spectral curve $y^2 = (x - 3)(x - 6)(x - 9)(x - 12)$ is birational over $\mathbb{Q}$ to the elliptic curve $E: y^2 = x^3 - x^2 - 4x + 4$, LMFDB label **24.a.4**, of conductor 24, with $j = 35152/9$ and torsion $\mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/4\mathbb{Z}$. The curve E is a Weierstrass model for the modular curve $X_0(24)$ and is *not* a CM curve. By the modularity theorem (Wiles–Breuil–Conrad–Diamond–Taylor) and the Eichler–Shimura relation, the Hecke eigenvalues $a_p(E) = p + 1 - \#E(\mathbb{F}_p)$ coincide with the Fourier coefficients of the unique newform $f \in S_2^{\text{new}}(\Gamma_0(24))$, LMFDB label **24.2.a.a**, which admits the eta-quotient expression $f(z) = \eta(2z)\eta(4z)\eta(6z)\eta(12z)$. A PARI/GP verification confirms the match $a_p(E) = a_p(f)$ at all primes $5 \leq p < 200$ with $p \nmid 24$ (44/44). We articulate this observation as an identification of two a priori unrelated spectra: the TEK eigenvalues of a pure lattice gauge theory matrix model and the Hecke eigenvalues of a weight-2 modular form, and we frame the follow-up question of whether the Hecke action extends to TEK Wilson operators at other levels.

1. INTRODUCTION

The twisted Eguchi–Kawai model. The Eguchi–Kawai (EK) reduction [7] replaces pure $SU(N)$ lattice gauge theory on a periodic d -dimensional torus by a one-site matrix model

$$S_{\text{EK}} = -\beta N \sum_{\mu < \nu} \Re \text{Tr}(U_\mu U_\nu U_\mu^\dagger U_\nu^\dagger), \quad U_\mu \in SU(N), \quad \mu = 1, \dots, d.$$

At leading order in $1/N$ and at strong coupling, this matrix model reproduces the full lattice theory. The reduction breaks at weak coupling because the $U(1)^d$ center symmetry breaks spontaneously. The twisted variant [8] restores center symmetry through a twist tensor $z_{\mu\nu} = \exp(2\pi i n_{\mu\nu}/N)$, antisymmetric and $\mathbb{Z}$ -valued; we work with the minimal symmetric twist $n_{\mu\nu} = \pm 1$ in $d = 4$ and $N = 3$, which is the simplest realization preserving center symmetry. Standard references for the construction, the twist eaters, and the resulting effective virtual lattice include [5, 2, 9].

Expanding the gauge field around the twist-eater background $U_\mu = \Gamma_\mu$, where the matrices $\Gamma_\mu \in SU(3)$ realize the Heisenberg–Weyl algebra $\Gamma_\mu \Gamma_\nu = z_{\mu\nu} \Gamma_\nu \Gamma_\mu$ (see (5) below), the quadratic part of the action gives a one-loop propagator

$$D^{-1}(p) = 4 \sum_{\mu=1}^4 \sin^2(p_\mu/2), \quad p_\mu = \frac{2\pi n_\mu}{\bar{L}}, \quad n_\mu \in \{0, 1, \dots, \bar{L} - 1\}, \quad \bar{L} = N = 3.$$

Since $\sin^2(\pi/3) = \sin^2(2\pi/3) = 3/4$, the eigenvalues $E(n_1, \dots, n_4) = 4 \sum_\mu \sin^2(\pi n_\mu/3)$ simplify to

$$(1) \quad E(n_1, \dots, n_4) = 3 \nu(n), \quad \nu(n) := \#\{\mu : n_\mu \neq 0\} \in \{1, 2, 3, 4\},$$

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so the spectrum collapses to the four-point set $\{3, 6, 9, 12\}$ with multiplicities $(8, 24, 32, 16)$ summing to $3^4 - 1 = 80$.

The spectral curve and main observation. The associated hyperelliptic spectral curve over $\mathbb{Q}$ is

$$(2) \quad C_{\text{TEK}}: y^2 = (x-3)(x-6)(x-9)(x-12).$$

With four simple rational branch points, C_{TEK} has genus one; it is therefore birational over $\mathbb{Q}$ to an elliptic curve, which we identify explicitly via the standard quartic-to-Weierstrass transformation (PARI/GP `ellfromeqn`) followed by `ellminimalmodel`:

$$(3) \quad E: y^2 = x^3 - x^2 - 4x + 4.$$

The curve E is LMFDB **24.a4**, Cremona **24A3**, with conductor $N(E) = 24$, discriminant $\Delta(E) = 2304 = 2^8 \cdot 3^2$, j -invariant $j(E) = 35152/9 = 2^4 \cdot 13^3/3^2$, torsion subgroup $E(\mathbb{Q})_{\text{tors}} \cong \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/4\mathbb{Z}$, and analytic and algebraic ranks both equal to 0 [10]. The denominator of $j(E)$ is 9, so $j(E)$ is *not* an algebraic integer; by a classical theorem of Deuring–Schneider (see, e.g., [15, Ch. II, Thm. 6.1]), E is *not* a CM elliptic curve.

By the modularity theorem of Wiles, Taylor–Wiles, and Breuil–Conrad–Diamond–Taylor [17, 16, 1], every elliptic curve over $\mathbb{Q}$ is modular: there is a unique normalized newform $f_E \in S_2^{\text{new}}(\Gamma_0(N(E)))$ such that $L(E, s) = L(f_E, s)$, and by Eichler–Shimura [13, Ch. 7] the Hecke eigenvalues $a_p(f_E)$ coincide with the trace of Frobenius $a_p(E) = p + 1 - \#E(\mathbb{F}_p)$ for every prime $p \nmid N(E)$. The space $S_2^{\text{new}}(\Gamma_0(24))$ is one-dimensional [10], spanned by the single newform

$$(4) \quad f(z) = \eta(2z)\eta(4z)\eta(6z)\eta(12z),$$

labeled **24.2.a.a** in the LMFDB.

Observation. The Hecke eigenvalues of the modular form $f = \mathbf{24.2.a.a}$ coincide with the trace of Frobenius of the spectral curve C_{TEK} of the $\text{SU}(3)$ -twisted Eguchi–Kawai model in $d = 4$ with the minimal symmetric twist. Explicitly, for every prime $p \nmid 24$,

$$a_p(C_{\text{TEK}}) := a_p(E) = a_p(f),$$

where E is the minimal model (3) of C_{TEK} .

This identifies the eigenvalue spectrum of a *lattice gauge theory matrix model* with the Hecke spectrum of a *weight-2 modular form*. The identification is a consequence of the modularity theorem applied to E and is not itself a new modularity statement; the novelty is that the elliptic curve in question arises canonically and unambiguously from a TEK matrix model construction.

Organization. Section 2 reviews the TEK construction at $\text{SU}(3)$ in $d = 4$ with the minimal symmetric twist and verifies the spectrum collapse (1). Section 3 describes the birational reduction of (2) to (3) and records the LMFDB invariants. Section 4 states and proves Theorem 3 and reports the PARI/GP verification at 44 primes $5 \leq p < 200$. Section 5 addresses non-CM status, the modular curve interpretation $X_0(24)$, and the relation to large- N limits. Section 6 formulates open questions about a Hecke action on TEK Wilson operators and the generalization to other levels and gauge groups.

Conventions. All elliptic curves are defined over $\mathbb{Q}$ unless stated otherwise. Newform conventions follow [10] and [11]. The PARI/GP version used for the numerical verification is 2.15.4 [12].

2. THE TEK PROPAGATOR SPECTRUM AT $SU(3)$

We work in $d = 4$ Euclidean dimensions, gauge group $SU(3)$, with the minimal symmetric flux $n_{\mu\nu} \in \{\pm 1\}$ satisfying $\sum_{\nu \neq \mu} n_{\mu\nu} \equiv 0 \pmod{3}$. The classical vacuum of the TEK action consists of the *twist eaters* $U_\mu = \Gamma_\mu$, solving

$$(5) \quad \Gamma_\mu \Gamma_\nu = z_{\mu\nu} \Gamma_\nu \Gamma_\mu, \quad z_{\mu\nu} = \exp(2\pi i n_{\mu\nu}/3),$$

which is a finite-dimensional realization of the Heisenberg–Weyl algebra at level 3. An explicit realization uses the Sylvester clock–shift pair

$$P = \text{diag}(1, \omega, \omega^2), \quad Q|j\rangle = |j+1 \bmod 3\rangle, \quad \omega = e^{2\pi i/3},$$

with $\Gamma_1 = P$, $\Gamma_2 = Q$, $\Gamma_3 = PQ$, $\Gamma_4 = P^2Q$ (up to relabeling). One verifies that the relations (5) hold for the chosen $n_{\mu\nu}$.

Expanding $U_\mu = e^{iaA_\mu} \Gamma_\mu$, with A_μ Hermitian and traceless and a the (formal) lattice spacing, the quadratic part of S_{TEK} gives a propagator acting on Fourier modes $A_\mu = \sum_n \tilde{A}_\mu(n) e^{ip \cdot x}$ with discrete momenta $p_\mu = 2\pi n_\mu / \bar{L}$ and effective lattice size $\bar{L} = N = 3$; see [5, Sec. 2] and [2, §3] for the derivation. The eigenvalues

$$E(n_1, n_2, n_3, n_4) = 4 \sum_{\mu=1}^4 \sin^2(\pi n_\mu/3), \quad n_\mu \in \{0, 1, 2\}, \quad n \neq 0,$$

collapse via $\sin^2(\pi/3) = \sin^2(2\pi/3) = 3/4$ to $E(n) = 3\nu(n)$, as in (1). The multiplicity of a given value $E = 3k$, $k = 1, 2, 3, 4$, is $\binom{4}{k} 2^k$, giving the count

$$(m_3, m_6, m_9, m_{12}) = (8, 24, 32, 16),$$

with total $8 + 24 + 32 + 16 = 80 = 3^4 - 1$, the full count of non-zero modes. This is verified numerically in PARI/GP (ancillary script `tek_verify.gp`, §1).

3. BIRATIONAL REDUCTION TO **24.a4**

The spectral curve (2) has four simple rational branch points, hence genus $g = 1$, and is therefore birational over $\mathbb{Q}$ to an elliptic curve. We give an explicit reduction.

Translating $x \mapsto x + 15/2$ symmetrizes the roots to $\{\pm 3/2, \pm 9/2\}$, so

$$y^2 = (x^2 - 9/4)(x^2 - 81/4) = x^4 - \frac{45}{2}x^2 + \frac{729}{16}.$$

Applying `ellfromeqn` (PARI/GP), which implements the classical quartic-to-Weierstrass map (see [3, §7.2] or [4, §8]) yields the (non-minimal) Weierstrass model W : $[a_1, a_2, a_3, a_4, a_6] = [0, 315, 0, 32724, 1\,122\,660]$, and `ellminimalmodel` produces

$$E: y^2 = x^3 - x^2 - 4x + 4.$$

PARI computes $\Delta(E) = 2304 = 2^8 \cdot 3^2$, $j(E) = 35152/9$, conductor $N(E) = 24 = 2^3 \cdot 3$, torsion $E(\mathbb{Q})_{\text{tors}} = \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/4\mathbb{Z}$, algebraic rank $\text{rk } E(\mathbb{Q}) = 0$; these match the LMFDB record for the curve **24.a4** [10].

Remark 1. Replacing $\bar{L} = 3$ with $\bar{L} = 2$ in the propagator (equivalently, rescaling momenta) yields branch points $\{4, 8, 12, 16\}$, which give the same cross-ratio $\lambda = 4/3$ and hence the same j -invariant; the resulting elliptic curve is $\mathbb{Q}$ -isomorphic to E . The structural identity between C_{TEK} and E is invariant under the discrete rescaling $\bar{L} \mapsto \bar{L}'$ so long as $\bar{L}$ is coprime to $N = 3$ and the symmetric twist is maintained. Generic asymmetric twists destroy the four-point collapse and produce higher-genus spectral curves.

Remark 2. The Weierstrass model E is in fact a model for the *modular curve* $X_0(24)$ [10]. The genus formula $g(X_0(N))$ (see [6, Thm. 3.1.1]) gives $g(X_0(24)) = 1$, and the dimension of new cusp forms of weight 2 at level 24 is $\dim S_2^{\text{new}}(\Gamma_0(24)) = 1$, so $J_0(24)$ is itself an elliptic curve and $X_0(24) \simeq J_0(24)$ over $\mathbb{Q}$. Thus the spectral curve C_{TEK} is itself a model for $X_0(24)$. We return to this in Section 5.

4. THE HECKE–FROBENIUS IDENTIFICATION

Theorem 3. *Let C_{TEK} be the TEK spectral curve (2) and let E be its minimal model (3). Let $f \in S_2^{\text{new}}(\Gamma_0(24))$ be the unique newform of level 24, weight 2, and trivial character, with q -expansion $f(z) = q \prod_{n \geq 1} (1 - q^{2n})(1 - q^{4n})(1 - q^{6n})(1 - q^{12n}) = \eta(2z)\eta(4z)\eta(6z)\eta(12z)$. Then for every prime $p \nmid 24$,*

$$a_p(E) = p + 1 - \#E(\mathbb{F}_p) = a_p(f).$$

Proof. The curve E of (3) has conductor 24; this is verified via Tate’s algorithm [14, Ch. IV] and recorded in the LMFDB as **24.a4**. By modularity [17, 16, 1], E corresponds to a unique normalized newform f_E of weight 2 and level $N(E) = 24$ with $L(E, s) = L(f_E, s)$. Since $\dim S_2^{\text{new}}(\Gamma_0(24)) = 1$ and the unique basis vector is the eta-quotient f of (4) [10], we must have $f_E = f$. Comparing local L -factors at unramified primes gives the Eichler–Shimura relation $a_p(E) = p + 1 - \#E(\mathbb{F}_p) = a_p(f)$ for $p \nmid 24$ [13, Ch. 7, Thm. 7.6]. $\square$

Proposition 4 (Numerical verification). *Let $\pi(x) = \#\{p \leq x : p \text{ prime}\}$. In a PARI/GP verification using the ancillary script `tek_verify.gp`, the equality $a_p(E) = a_p(f)$ was checked exhaustively for the 44 primes p with $5 \leq p < 200$ and $p \nmid 24$, with full agreement (44/44 match, no exceptions). The first fifteen values are recorded in Table 1.*

TABLE 1. The first fifteen Hecke / Frobenius eigenvalues $a_p(E) = a_p(f)$ for $p \nmid 24$. All values are integers in the Hasse interval $|a_p| \leq 2\sqrt{p}$. Source: PARI/GP `ellap` and `mfcoef`.

p	$a_p(E)$	$a_p(f)$	p	$a_p(E)$	$a_p(f)$	p	$a_p(E)$	$a_p(f)$
5	−2	−2	17	2	2	37	6	6
7	0	0	19	−4	−4	41	−6	−6
11	4	4	23	−8	−8	43	4	4
13	−2	−2	29	6	6	47	0	0
			31	8	8			

Remark 5. We emphasize the modest scope of Theorem 3: it is not a new modularity result. The statement is, in effect, a triple identification:

- (1) the TEK propagator spectrum at $\text{SU}(3)$, $d = 4$, minimal symmetric twist gives the curve C_{TEK} of (2);
- (2) the minimal model of C_{TEK} is the elliptic curve **24.a4**;
- (3) the modularity theorem then forces the trace-of-Frobenius spectrum of C_{TEK} to match the Hecke eigenvalues of the unique level-24 weight-2 newform.

The content of the present note lies in steps (1)–(2): a clean and canonical pure-gauge-theoretic construction whose minimal model hits exactly the modular curve $X_0(24)$. Step (3) is then a classical consequence of modularity. The further question – whether the Hecke operators T_p act in a natural way on TEK Wilson observables – is left open.

Remark 6 (Non-vacuum flux sector). The construction of Section 2 uses the minimal symmetric twist $k = 1$ of the González-Arroyo-Okawa Eguchi-Kawai matrix model. On the four-torus T^4 , this twist corresponds to a non-trivial class in $H^2(T^4, \mathbb{Z}_3) \cong (\mathbb{Z}_3)^6$, explicitly carrying Pfaffian $+1 \pmod{3}$ in

the canonical six-component flux basis. The Heisenberg-Weyl realisation of Section 2 is intrinsically tied to this non-vacuum flux sector: replacing $k = 1$ by $k = 0$ collapses the spectral curve to the trivial point. Consequently Theorem 3 is a statement about TEK in a *specific superselection sector* of the lattice gauge bundle; it does not directly engage the vacuum-sector mass-gap question of the Clay Yang-Mills Millennium problem. The suggestive factorisation $N(E) = 24 = 8 \cdot 3$ into a power of two times the rank of the centre $\mathbb{Z}_3$ is not part of a broader pattern: the analogous $SU(2)$ construction would require a conductor-six elliptic curve, of which none exists in LMFDB. We therefore present the present result as a coincidence of arithmetic structure with the minimal twisted reduction, *not* as a general functor $\text{TEK} \mapsto \text{HeckeAlg}$ of the kind discussed in Section 6.

5. REMARKS

5.1. Non-CM and arithmetic non-overlap with Heegner anchors. The denominator of $j(E) = 35152/9$ is 9, so $j(E)$ is not an algebraic integer; E is therefore not a CM elliptic curve. Equivalently, the conductor $N(E) = 24$ does not appear in the class-number-one list of conductors of CM elliptic curves over $\mathbb{Q}$, which is $\{27, 32, 36, 49, 121, 256, 361, 1849, 4489, 26569\}$ attached to $K = \mathbb{Q}(\sqrt{-D})$ with $D \in \{3, 4, 7, 8, 11, 19, 43, 67, 163\}$ [15, App. A, Thm. A.14]. A statistical confirmation is obtained from the Hecke density of supersingular primes: the number of primes $p \in [5, 2000] \setminus \{2, 3\}$ with $a_p(E) = 0$ is $7/301 \approx 2.3\%$, consistent with the Lang–Trotter conjecture for non-CM curves (heuristic density $O(p^{-1/2})$) and incompatible with the 50% density of a CM curve at primes inert in the CM field.

5.2. The modular curve interpretation. By Remark 2, the TEK spectral curve C_{TEK} is a Weierstrass model for the modular curve $X_0(24)$. Hence the underlying gauge-theoretic data of the TEK matrix model at $SU(3)$, $d = 4$ is naturally encoded in $X_0(24)$, and the Hecke algebra $\mathbb{T}_{24}$ of $S_2(\Gamma_0(24))$ acts on the cotangent space of $X_0(24)$. The Hecke operators T_p for $p \nmid 24$ therefore act on the differential $\omega_f = f(z) dz$, with eigenvalues $a_p(f) = a_p(E)$. The natural follow-up question (Section 6) is whether this action lifts to a Hecke action on gauge-invariant observables of the TEK matrix model itself.

5.3. Robustness under $\bar{L}$ variation. For $\bar{L} = 2$ (equivalently, alternate momentum normalization), the branch points become $\{4, 8, 12, 16\}$, with the same cross-ratio $\lambda = 4/3$ and the same j -invariant; the resulting curve is $\mathbb{Q}$ -isomorphic to **24.a4**. The identification with **24.a4** is thus invariant under the discrete rescaling $\bar{L} \mapsto \bar{L}'$ so long as $\bar{L}$ is coprime to $N = 3$. Asymmetric or non-minimal twists destroy the four-point spectrum and produce higher-genus spectral curves; see [5, 2] for the twist-dependence of the spectrum.

5.4. Relation to large- N limits. The TEK construction is most useful as a finite- N tool for extracting large- N expectation values; in this note, we have focused on the $SU(3)$ case where the spectrum collapses exactly, and the spectral curve is the finite-dimensional matrix model spectrum, not an $N \rightarrow \infty$ limit. For $SU(N)$ with larger N , $\bar{L}$ coprime to N , the analogous propagator $E(n) = 4 \sum_{\mu} \sin^2(\pi n_{\mu}/\bar{L})$ does not collapse to a four-point set; the analog of Observation 1 at higher N is open and is one of the questions we record below.

6. OPEN QUESTIONS

Question 7. Does the Hecke action of $\mathbb{T}_{24}$ on $H^0(X_0(24), \Omega^1) = \mathbb{C} \cdot \omega_f$ extend to a natural action on gauge-invariant observables of the $SU(3)$ -TEK matrix model? Concretely, is there an operator $\hat{T}_p$ on the algebra of $SU(3)$ -invariant polynomials in the matrices $U_1, \dots, U_4$ whose restriction to the character ring of Wilson loops matches the action of T_p on ω_f ?

Question 8. For other gauge groups $SU(N)$ with twist parameters giving elliptic spectral curves of conductor M , does the analog of Observation 1 hold? In particular, is there a finite list of $(N, d, n_{\mu\nu})$ for which the TEK spectral curve is $\mathbb{Q}$ -rationally equivalent to a modular curve $X_0(M)$ of small conductor M ?

Question 9. Define a finite- N analog of the trace of Frobenius: $\tilde{a}_p(\text{TEK}_N) := p + 1 - \#C_{\text{TEK}}(\mathbb{F}_p)$ for the spectral curve C_{TEK} at gauge group $SU(N)$. For which sequences $N \rightarrow \infty$ along which the spectral curve remains elliptic, does $\tilde{a}_p(\text{TEK}_N) \rightarrow a_p(f)$ for some weight-2 newform f ? This is a quantitative form of Question 8.

Question 10. Is there a functor $F: \text{TEK}_{\text{twist,ell.}} \rightarrow \text{HeckeAlg}_{S_2^{\text{new}}}$ from the category of *TEK twist data whose spectral curve is elliptic* to the category of Hecke algebras of weight-2 newforms, such that $F(\text{TEK-SU}(3)\text{-min. twist}) = \mathbb{T}_{24}$? If so, what is its image, and what are its kernel and cokernel? Even a partial answer (e.g. extending to one or two other gauge-theoretic data) would constitute a structural step beyond the present *ad hoc* identification.

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The author thanks the LMFDB collaboration [10] for the public database that made this note possible, and the developers of PARI/GP [12] for the verification tool. Numerical verifications were performed on a Hostinger VPS; the ancillary script `tek_verify.gp` is provided alongside this manuscript and reproduces all numerical claims (Section 2, Section 3, and Proposition 4) in under five seconds on standard hardware.

Reproducibility. The PARI/GP verification script `tek_verify.gp` accompanying this note executes the following checks:

- (1) Compute the multiplicities $(m_3, m_6, m_9, m_{12}) = (8, 24, 32, 16)$ of the TEK propagator spectrum directly from (1).
- (2) Reduce C_{TEK} to Weierstrass form via `ellfromeqn` and `ellminimalmodel`, yielding E of (3).
- (3) Record $\text{cond}(E) = 24$, $\Delta(E) = 2304$, $j(E) = 35152/9$, $E(\mathbb{Q})_{\text{tors}} = \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/4\mathbb{Z}$.
- (4) Confirm $j(E) \notin \mathbb{Z}$, hence E is not CM.
- (5) Load the unique newform $f \in S_2^{\text{new}}(\Gamma_0(24))$ via `mfininit/mfeigenbasis`, and verify the equality $a_p(E) = a_p(f)$ for all primes $5 \leq p < 200$ with $p \nmid 24$; the script reports 44/44 matches.
- (6) Verify the supersingular density: $7/301 \approx 2.3\%$ for primes $p \in [5, 2000]$, consistent with the non-CM case.

APPENDIX A. THE VERIFICATION SCRIPT `TEK_VERIFY.GP`

The file `tek_verify.gp` is included as an ancillary file. Its body is reproduced here for self-containment.

```
/* tek_verify.gp -- ancillary file for Paper_TEK_X024_Note
 * Verification script for: TEK SU(3) spectral curve = E[24.a4] = 24.2.a.a
 *
 * Reproducibility: PARI/GP 2.15.4
 * Run:  gp -q < tek_verify.gp
 */

/* 1. TEK SU(3) propagator spectrum at one-loop around twist eaters */
/*   E(n) = 4 * sum_{mu=1..4} sin^2(pi * n_mu / 3), n_mu in {0,1,2}, n != 0 */
print("=== 1. TEK SU(3) d=4 minimal symmetric twist propagator spectrum ===");
\\ E(n) = 4 sum_mu sin^2(pi n_mu / 3) for n_mu in {0,1,2}, n != 0
```

```

\\ Since sin^2(pi/3) = sin^2(2pi/3) = 3/4, E = 3 * (number of nonzero n_mu)
\\ so the spectrum is {3, 6, 9, 12}.
\\ tally pass
tek_tally() = {
  my(mult_3 = 0, mult_6 = 0, mult_9 = 0, mult_12 = 0, nz);
  for(a = 0, 2, for(b = 0, 2, for(c = 0, 2, for(d = 0, 2,
    if(a + b + c + d > 0,
      nz = (a != 0) + (b != 0) + (c != 0) + (d != 0);
      if(nz == 1, mult_3 += 1);
      if(nz == 2, mult_6 += 1);
      if(nz == 3, mult_9 += 1);
      if(nz == 4, mult_12 += 1);
    )
  ))));
  print("Multiplicities: E=3 -> ", mult_3, "; E=6 -> ", mult_6, "; E=9 -> ", mult_9, "; E=12
  print("Total non-zero modes: ", mult_3+mult_6+mult_9+mult_12, " (expected 3^4 - 1 = 80)");
};
tek_tally();

/* 2. Spectral curve y^2 = (x-3)(x-6)(x-9)(x-12) -> Weierstrass form */
print("\n=== 2. Spectral curve birational reduction ===");
C = ellfromeqn(y^2 - (x-3)*(x-6)*(x-9)*(x-12));
print("ellfromeqn output: ", C);
EW = ellinit(C);
Emin = ellminimalmodel(EW);
print("Minimal model: ", Emin[1..5]);

/* 3. Verify E[24.a4] : y^2 = x^3 - x^2 - 4x + 4 */
print("\n=== 3. Curve E[24.a4] : y^2 = x^3 - x^2 - 4x + 4 ===");
E = ellinit([0, -1, 0, -4, 4]);
print("Conductor : ", ellglobalred(E)[1]);
print("j-invariant : ", E.j, " (= ", factor(numerator(E.j)), " / ", factor(denominator(E.j)), "
print("Discriminant : ", E.disc, " (= ", factor(E.disc), " )");
print("Torsion (over Q) : ", elltors(E)[1]);

/* 4. NON-CM check: j has denominator 9 */
print("\n=== 4. NON-CM check (Schneider-Deuring) ===");
print("Denominator of j = ", denominator(E.j), " != 1");
print("j NOT algebraic integer => E NOT CM");

/* 5. Modular form 24.2.a.a a_p match */
print("\n=== 5. Hecke a_p match: E vs f_{24.2.a.a} ===");
mf = mfinit([24, 2, 1], 0);
L = mfeigenbasis(mf);
print("Number of newforms at level 24, weight 2: ", #L);
f = L[1];

check_match(pmax) = {
  my(c = 0, t = 0, mm = []);
  forprime(p = 5, pmax,

```

```

    t += 1;
    my(ap_E = ellap(E, p), ap_f = mfcoef(f, p));
    if(ap_E == ap_f, c += 1, mm = concat(mm, [[p, ap_E, ap_f]]));
);
print("Match: ", c, "/", t);
if(#mm == 0, print("ALL MATCH"), print("Mismatch: ", mm));
};
check_match(199);

/* 6. Supersingular density (non-CM gives ~0, CM gives ~1/2) */
print("\n=== 6. Supersingular density (CM marker) ===");
ss_density(pmax) = {
    my(c = 0, t = 0);
    forprime(p = 5, pmax,
        t += 1;
        if(ellap(E, p) == 0, c += 1);
    );
    print("Primes 5..", pmax, ": ", c, "/", t, " = ", c*1.0/t);
};
ss_density(200);
ss_density(2000);

/* 7. Print a_p table for primes 5..50 */
print("\n=== 7. a_p table for primes 5..50 ===");
my_table() = {
    forprime(p = 5, 50,
        print("p=", p, " a_p=", ellap(E, p));
    );
};
my_table();

```

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